

The Synthetic Control Method

Causal effects of interventions on a single aggregate unit

ECO372H1F — Data Analysis & Applied Econometrics in Practice

Where we are, and where we are going

We have built a toolkit for causal inference with *many* units:

- **DiD / panel methods** — many treated and control units, parallel trends.
- **RDD** — many units sorted around a cutoff.
- **IV** — many units, an instrument breaking endogeneity.

Today: what if you have essentially *one* treated unit?

- One state passes a law. One country reunifies. One firm changes a price.
- There is no natural “control state” and no large sample of treated cases to average over.

The **synthetic control method (SCM)** builds a control unit *from data* — a weighted average of untreated units that mimics the treated unit before the intervention.

1. **Motivation & applications** — why the method exists; business and policy uses.
2. **The basic SCM** (the core of the lecture)
 - Setup, the estimator, predictor weights
 - Why SCM beats regression: the four advantages
 - The factor model and the bias bound (intuition)
 - Contextual and data requirements
 - Inference and diagnostics
 - Robustness checks
3. **Extensions** — penalized SC, augmented SC, synthetic DiD, staggered adoption, and the A/B-testing link.

Primary running example: **California's Proposition 99** tobacco-control program. Secondary: **German reunification** (for the regression comparison and robustness).

Motivation & Applications

The problem SCM was built for

Many interventions we care about happen at an **aggregate level**, to a **small number of large units**:

- a tax reform in one state, a peace process in one region, a currency peg in one country;
- a pricing or feature change rolled out to one market, client, or product line.

Why the usual machinery struggles:

- **Large- N regression** needs many observed instances of the event — we have one.
- **Single-unit time series** can pick up short, sharp effects but is swamped by other shocks over the medium/long run.
- **Comparative case studies** (pick one “similar” unit) work *if* you can defend the comparison — but the choice is informal and contestable.

Even in experiments, micro-randomization may be infeasible (fairness, legality) or ineffective (interference/spillovers).

The business framing: SCM vs. A/B testing

The gold standard for causal questions is the **randomized experiment** / **A/B test**: randomly split units into treatment and control; any post-difference is causal.

But A/B tests are often **infeasible or expensive**:

- the change affects an *entire* group (a market, a publisher, a country) — no within-group randomization;
- ethical, legal, or technical barriers to randomizing;
- the “experiment” is a passive external shock that already happened.

SCM is the workhorse alternative: instead of a *randomized* control group, construct a *synthetic* one from units that were not treated.

Widely used in business analytics units (e.g. Uber, Microsoft/Xandr) precisely because it needs no random assignment — only similar units and pre-period history.

Two naive approaches that fail (and why)

1. **Before vs. after** (compare the treated unit to its own past):

- Confounded by anything else changing over time — **seasonality**, trends, macro shocks.
- Example: ad-platform revenue has strong weekly/quarterly seasonality; a post-change dip could be the calendar, not the change.

2. **One hand-picked comparison unit** (treated vs. a single “similar” control):

- Which unit? The choice is itself a research-design decision that drives the answer.
- No single unit may track the treated unit well — different units follow different dynamics.

SCM's answer: don't pick one control — let the data choose a **weighted combination** of many that reproduces the treated unit's pre-period.

The core idea of SCM

Synthetic control

A **weighted average of untreated units** (the *donor pool*) whose weights are chosen so that the synthetic unit closely matches the treated unit's **characteristics and outcome path before the intervention**.

- If the synthetic unit tracks the treated unit *before* treatment, it is a credible stand-in for what would have happened *after*, absent treatment.
- The causal effect is read off as the **gap** between the treated unit and its synthetic control in the post-period.
- A combination of units typically reproduces the treated unit far better than any single unit alone.

The method formalizes comparative case studies: the control is a transparent, data-driven *construction*, not a judgment call.

Applications I — business & industry

SCM is heavily used outside academia (consulting, tech, multilateral orgs, governments).

Examples:

- **Ad tech (Xandr/Microsoft).** Effect of changing a bid-request field on revenue for one ad placement; synthetic control built from “similar” placements (donor pool), since a full A/B test is costly.
- **Feature rollouts.** Launch a feature to one client/market first; estimate its effect against a synthetic counterpart of the others.
- **Marketing.** Effect of offline TV advertising on online chatter; effect of a mobile-hailing technology on drivers’ hourly earnings.
- **Policy-as-business.** Mandatory GMO labeling (Vermont) on demand; an “open payments” disclosure law on physician prescribing.

Signature pattern: one treated unit, several donor candidates, one sharp moment of intervention.

Applications II — economics & policy

- **Tobacco control** — California's Proposition 99 (Abadie, Diamond, Hainmueller 2010).
Our running example.
- **German reunification** — effect on West German per-capita GDP (ADH 2015).
- **Terrorism** — economic cost of conflict in the Basque Country (Abadie & Gardeazabal 2003), the original SCM paper.
- **Right-to-carry laws** and crime; **legalized prostitution** and public health; **immigration** policy; **taxation** of high earners; **organized crime**.
- Both sides of the **minimum-wage** and **immigration** debates have used SCM.
- Used well beyond economics: political science, biomedical research (e.g. child-mortality effects of democratic reforms), engineering.

Athey & Imbens called SCM “arguably the most important innovation in the policy evaluation literature in the last 15 years.”

From informal case studies to SCM

Classic comparative case studies chose a comparison *informally*:

- **Mariel boatlift** (Card 1990): Miami vs. four other US cities.
- **Card & Krueger (1994)**: New Jersey vs. Pennsylvania for a minimum-wage change.

The weakness: the comparison rests on an **informal claim of similarity**, and with a few aggregate units, *no single unit* may be a good match.

SCM keeps the case-study spirit but adds:

- a **formal, data-driven rule** for building the comparison, and
- a **mode of inference** that the formal rule makes possible (placebo/permutation tests — later).

The Basic Synthetic Control Method

Setup and notation

- $J + 1$ units, indexed $j = 1, \dots, J + 1$, observed for T periods $t = 1, \dots, T$.
- Unit $j = 1$ is the **treated** unit; it is exposed from period $T_0 + 1$ onward.
- Units $j = 2, \dots, J + 1$ are the **donor pool**: untreated throughout.
- For each unit and period we observe the outcome Y_{jt} .
- For each unit we observe k predictors $X_{1j}, \dots, X_{kj}$ (may include pre-period outcomes), *unaffected by treatment*.

Collect predictors:

$$X_1 = \begin{pmatrix} X_{11} \\ \vdots \\ X_{k1} \end{pmatrix} (k \times 1), \quad X_0 = [X_2 \ \cdots \ X_{J+1}] (k \times J).$$

X_1 = treated unit's characteristics; X_0 = same characteristics for every donor.

Potential outcomes and the target parameter

Using the potential-outcomes framework:

- Y_{jt}^N — outcome *without* intervention.
- Y_{1t}^I — outcome *with* intervention (treated unit, $t > T_0$).

The effect on the treated unit at time $t > T_0$:

$$\tau_{1t} = Y_{1t}^I - Y_{1t}^N = Y_{1t} - Y_{1t}^N.$$

because for $t > T_0$ we *observe* the treated outcome, $Y_{1t} = Y_{1t}^I$.

The whole problem in one line

Y_{1t} is observed; Y_{1t}^N is the **counterfactual** we never see. Estimating the effect = estimating Y_{1t}^N for $t > T_0$.

Note: τ_{1t} is allowed to *vary over time* — effects can build up or fade. No constant-effect restriction.

The estimator

A synthetic control is a weight vector $W = (w_2, \dots, w_{J+1})'$. Given W :

$$\hat{Y}_{1t}^N = \sum_{j=2}^{J+1} w_j Y_{jt}, \quad \hat{\tau}_{1t} = Y_{1t} - \hat{Y}_{1t}^N.$$

Weights are restricted:

$$w_j \geq 0 \text{ for all } j, \quad \sum_{j=2}^{J+1} w_j = 1.$$

So the synthetic control is a **convex combination** — a proper weighted average of the donors.

Caveat: weights summing to one makes sense when variables are scaled to remove size differences (per-capita, prices). If a variable scales with unit size, rescale first.

Why restrict the weights?

The constraints $w_j \geq 0$, $\sum w_j = 1$ are the heart of SCM. They deliver:

- **No extrapolation.** The synthetic unit stays inside the *convex hull* of the donors — it can never be more extreme than the data we actually have.
- **Sparsity.** Typically only a handful of donors get nonzero weight (more on this soon).
- **Interpretability.** The counterfactual is a transparent average of identifiable real units.

Relaxing them (allowing negative weights, or weights > 1) buys flexibility but allows extrapolation — and, as we'll see, *that is exactly what linear regression does*.

Choosing W : match the pre-intervention characteristics

Pick W^* to make the synthetic unit resemble the treated unit *before* treatment:

$$W^* = \arg \min_W \|X_1 - X_0 W\| \quad \text{s.t.} \quad w_j \geq 0, \sum_j w_j = 1.$$

i.e. choose weights so that, predictor by predictor,

$$X_{h1} \approx w_2 X_{h2} + \cdots + w_{J+1} X_{h,J+1}, \quad h = 1, \dots, k.$$

- X_1, X_0 usually include **pre-period outcomes** plus other predictors of the outcome.
- We minimize the discrepancy on *characteristics*; the payoff is tracking the *outcome path*.

The predictor weights v_h

Not all predictors matter equally. The norm is a **weighted** distance:

$$\|X_1 - X_0 W\|_V = \left(\sum_{h=1}^k v_h (X_{h1} - w_2 X_{h2} - \dots - w_{J+1} X_{h,J+1})^2 \right)^{1/2}.$$

- $v_1, \dots, v_k \geq 0$ reflect how predictive each characteristic is of the counterfactual outcome Y_{1t}^N .
- Chosen by the analyst, or **data-driven**: pick V so the resulting $W^*(V)$ best fits the pre-period *outcomes* (often via cross-validation on a hold-out pre-period).

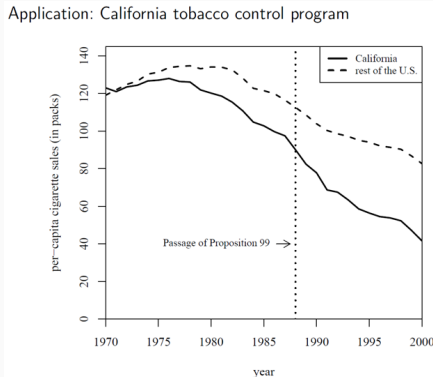
This is a **nested optimization**: outer loop over V (predictor importance), inner loop over W (donor weights). Different software defaults exist — report what you used.

Running example: California's Proposition 99

- In 1988 California passed **Proposition 99**: a 25-cent/pack cigarette tax, with funds for anti-smoking education and enforcement.
- **Outcome**: per-capita cigarette sales (packs).
- **Treated unit**: California. **Donor pool**: other US states.
- **Predictors**: retail price, per-capita income, share of population aged 15–24, per-capita beer consumption, and several pre-1988 values of cigarette sales.

Donor-pool hygiene (we'll formalize later): drop states that ran their own large tobacco programs or raised tobacco taxes in the sample period — they are “contaminated” controls.

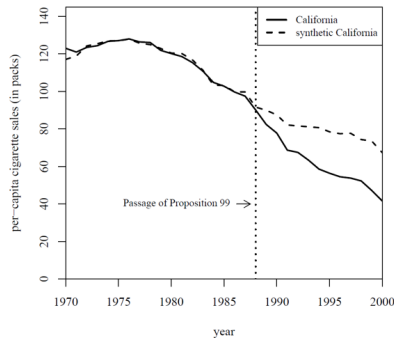
California vs. the rest of the US (raw)



The rest of the US is a poor comparison: California already smoked less and was trending differently. A single aggregate “other states” line will not do.

California vs. synthetic California

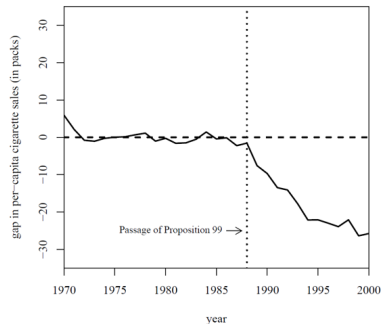
Application: California tobacco control program



The synthetic control (a weighted mix of a few states) tracks California *closely before 1988*, then the two diverge — the gap is the estimated effect.

The treatment effect = the gap

Application: California tobacco control program



$\hat{\tau}_{1t} = Y_{1t} - \hat{Y}_{1t}^N$. Roughly zero before 1988 (good fit), then a widening negative gap: Prop 99 reduced sales, with the effect growing for about a decade.

Reading the fit: predictor balance (German reunification)

ADH (2015) estimate the effect of the 1990 reunification on West German per-capita GDP.

Balance table:

	West Germany	Synthetic West Germany	OECD sample avg.
GDP per capita	15808.9	15802.2	13669.4
Trade openness	56.8	56.9	59.8
Inflation rate	2.6	3.5	7.6
Industry share	34.5	34.5	34.0
Schooling	55.5	55.2	38.7
Investment rate	27.0	27.0	25.9

Column 1 = X_1 , column 2 = $X_0 W^*$, column 3 = simple average of the 16 donor countries. The synthetic control matches column 1 **far better** than the raw average does — except inflation, where West Germany is near-extreme.

Reading the counterfactual: the weights

The synthetic West Germany is built from just a few countries:

Country	w_j^*	Country	w_j^*
Austria	0.42	Netherlands	0.09
United States	0.22	Switzerland	0.11
Japan	0.16	all others	0

- Five of sixteen donors carry all the weight; the rest are exactly zero — this is **sparsity**.
- You can *name* the counterfactual: “West Germany $\approx$ 42% Austria + 22% US + 16% Japan + ...”
- This transparency lets domain experts judge the comparison and reason about bias direction (we return to this).

Why use SCM? Compare it to linear regression

A regression counterfactual also turns out to be a weighted average of donor outcomes:

$$\hat{Y}_{1t}^{N,\text{reg}} = \hat{B}' \bar{X}_1 = Y_0 W^{\text{reg}}, \quad W^{\text{reg}} = \bar{X}_0'(\bar{X}_0\bar{X}_0')^{-1}\bar{X}_1,$$

where $\bar{X}$ augments X with a row of ones, and Y_0 is the matrix of donor post-period outcomes.

- Like SCM, the regression weights W^{reg} **sum to one**.
- **Unlike** SCM, they need not lie in $[0, 1]$ — they can be negative or exceed one.
- So regression is “a synthetic control that is allowed to extrapolate” and is *not* sparse.

SCM = regression + the constraints $w_j \geq 0$ and $\sum w_j = 1$. Those constraints are the whole story.

Regression weights extrapolate (German reunification)

The regression counterfactual for West Germany uses **negative** weights:

Country	W_j^{reg}	Country	W_j^{reg}
Australia	0.12	Netherlands	0.14
Austria	0.26	New Zealand	0.12
Belgium	0.00	Norway	0.04
Denmark	0.08	Portugal	-0.08
France	0.04	Spain	-0.01
Greece	-0.09	Switzerland	0.05
Italy	-0.05	United Kingdom	0.06
Japan	0.19	United States	0.13

Four countries enter with negative weight. The weights are dense (almost every country is used) and impossible to interpret as a “comparison group.”

Advantage 1 — No extrapolation

- Convex weights keep the synthetic unit inside the **convex hull** of the donors.
- Regression guarantees a *perfect* fit on characteristics ($\bar{X}_0 W^{\text{reg}} = \bar{X}_1$) **even when no donor resembles the treated unit** — by extrapolating outside the data.
- SCM refuses to do this. If the treated unit is far from every donor, SCM *shows* you the residual discrepancy rather than papering over it.

Extrapolation-based counterfactuals can be wildly off the support of the data — a known danger (King & Zeng 2006).

Advantage 2 — Transparency of the fit

- SCM makes the discrepancy $X_1 - X_0 W^*$ **visible** — it is the gap between columns 1 and 2 of the balance table.
- You can see, predictor by predictor, how well (or badly) the donors approximate the treated unit *by interpolation only*.
- If that table reveals you *cannot* reproduce the treated unit's characteristics from any weighted average, ADH advise: **do not use SCM**.

Regression hides this: it always reports a perfect characteristic-fit, so a bad comparison is invisible.

Advantage 3 — Safeguard against specification search

- SCM weights are computed from **pre-treatment data only** — no post-treatment outcomes enter the design phase.
- So *all* design choices (donor pool, predictors, weights) can be made **without knowing how they affect the result**.
- Weights can even be **pre-registered** before outcomes are realized — a pre-analysis plan for observational work.

Why this matters

This structurally blocks p -hacking and “specification searches”: you cannot tune the design to get the answer you want, because the answer isn’t visible while you design.

Advantage 4 — Transparency of the counterfactual & bias reasoning

Because weights are sparse and interpretable, you can reason about **the direction of bias**.

Example (German reunification): Austria, the Netherlands, Switzerland — small neighbours — carry large weight.

- If reunification *hurt* these neighbours (West Germany diverted demand/investment toward East Germany)...
- ...then their GDP is artificially low, so the synthetic counterfactual is too low...
- ...and the estimated negative effect is **attenuated** — a *lower bound* on the true magnitude.

You can only make this argument because you know *which* units compose the counterfactual and *by how much*. Dense, signed regression weights make it hopeless.

Advantage 5 — Sparsity (and why it differs from the lasso)

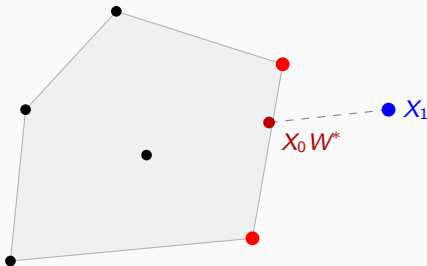
- SCM weights are typically **sparse**: most donors get exactly zero weight.
- This is a *geometric* consequence of projecting onto a convex hull, not a penalty we added.

Contrast with the lasso:

- Lasso sparsity is a *regularization device* to prevent overfitting; the identity of nonzero coefficients is usually not the point.
- SCM sparsity is *informational*: the identity and size of nonzero weights tell you what the counterfactual *is* and let you evaluate its validity.

Both aim at out-of-sample prediction (here: predicting Y_{1t}^N after T_0) — but only SCM treats the weights themselves as the object of interpretation.

Sparsity — the geometric picture



- Black dots: donor characteristics (columns of X_0). Shaded region: their convex hull.
- If X_1 lies **outside** the hull, $X_0 W^*$ is its unique projection onto the hull.
- Only the donors defining that face get weight (red); the rest are zero $\Rightarrow$ sparse.
- Number of nonzero weights is bounded by the number of predictors k .

When sparsity/uniqueness can fail

If X_1 lies *inside* the convex hull of the donors:

- there may be **infinitely many** weight vectors that achieve a perfect fit;
- the minimizer need not be unique or sparse ...
- ... though a sparse solution (with $\leq k + 1$ nonzero weights) *always* exists, by Carathéodory's theorem.

With a single treated unit and even moderate k , the **curse of dimensionality** usually keeps X_1 *outside* the hull, so this is rarely a problem.

It becomes a real issue with *many* treated units — the penalized estimator (later) is one fix.

What has to be true? The linear factor model

SCM's properties are studied under a **linear factor (interactive fixed effects) model** for the no-treatment outcome:

$$Y_{it}^N = \delta_t + \theta_t Z_i + \lambda_t \mu_i + \varepsilon_{it}.$$

- δ_t : common time trend.
- Z_i : *observed* unit features; θ_t time-varying returns to them.
- μ_i : **unobserved** unit features (factor loadings); λ_t common factors.
- ε_{it} : transitory unit-level shock (noise), mean zero.

This generalizes DiD: DiD assumes $\lambda_t \mu_i$ is additively separable into unit + time effects (a single, time-constant factor). SCM allows the unobserved factors to load differently over time.

Why matching the pre-period works

Suppose we find weights that reproduce the treated unit's observed features *and* its pre-period outcomes:

$$\sum_j w_j^* Z_j = Z_1, \quad \sum_j w_j^* Y_{j1} = Y_{11}, \dots, \sum_j w_j^* Y_{jT_0} = Y_{1T_0}.$$

- Matching T_0 pre-period outcomes (which load on $\lambda_t \mu_i$) forces the synthetic control to match the **unobserved** factor loadings μ_1 as well — *if* the pre-period is long enough.
- Then the synthetic control reproduces Y_{1t}^N after T_0 , up to noise.

Intuition: a unit that walked in lockstep with California for 20 years, across many shocks, probably shares California's hidden structure — not just by luck.

The bias bound — intuition

ADH (2010) bound the bias of $\hat{\tau}_{1t}$ under the factor model. The exact inequality is technical; the message is simple:

What the bound says

Bias is **small** when the pre-period fit is close *and* the number of pre-periods T_0 is large relative to the scale of the noise ε_{it} .

Roughly, $|\text{bias}| \propto \frac{(\text{noise scale})}{T_0^c}$ for some $c > 0$, *provided* the synthetic control fits the pre-period closely.

- More pre-periods $\Rightarrow$ harder to fit by luck $\Rightarrow$ smaller bias.
- Noisier outcomes $\Rightarrow$ easier to fit by luck $\Rightarrow$ bigger overfitting risk.

Practical implications of the bound

- **Close pre-period fit is the credibility test.** A synthetic control that does *not* track the treated unit before treatment is not to be trusted — there are *no ex-ante guarantees* of fit.
- **If the fit is poor, ADH recommend against using SCM** (or transforming the problem).
- **Overfitting risk** is high when T_0 is small, J (donors) is large, and noise is large — you can fit the pre-period perfectly and still predict the counterfactual badly.
- **Restrict the donor pool** to units genuinely similar to the treated unit, to limit interpolation bias and overfitting.

Contextual requirements (1): effect size vs. volatility

SCM (like any comparative case study) estimates the effect on *one* unit — there is no averaging to cancel noise.

- A **small** effect is indistinguishable from ordinary shocks to the outcome.
- Even a *large* effect can be undetectable if the outcome is *highly volatile*.
- Trouble comes only from **unit-specific** volatility (ε_{jt}). Volatility from *common* factors is differenced out by a good synthetic control.

Fixes: filter/de-noise the outcome (treated unit *and* donors) before fitting; prefer outcomes where the expected effect is large relative to noise.

This is why SCM has been most successful on smooth aggregates (GDP, mortality, cigarette sales) rather than noisy daily metrics.

Contextual requirements (2): a clean comparison group

The donor pool must actually contain valid controls:

- **Not everyone can be treated.** Drop units that adopted a *similar* intervention during the study — they are not untreated.
- **Drop big idiosyncratic shocks.** Remove donors that suffered large unrelated shocks that wouldn't have hit the treated unit.
- **Keep donors similar.** Even with no extrapolation, **interpolation bias** arises if the fit averages away large discrepancies between very different units.

Tobacco example: states with their own large tobacco programs / tax hikes are discarded. German reunification: restrict to OECD economies.

Contextual requirements (3): no anticipation

If agents react *before* the formal intervention — or parts of it are implemented early — the pre-period is contaminated and SCM is biased.

Fix: backdate the “treatment date” in the data to before any anticipation could begin.

- This does *not* mechanically bias the estimate: SCM lets the effect vary over time, so periods truly unaffected simply show ≈ 0 .
- Genuine effects then appear only when the real intervention bites.

Contrast with restrictive panel models that force a constant post-period effect — there, mislabeling periods *would* bias things.

Contextual requirements (4): no interference (SUTVA)

Potential outcomes are defined by a unit's *own* treatment only — no **spillovers** across units.

- If the treatment spills onto donors, those donors are bad counterfactuals.

A genuine **tension**:

- You *want* donors hit by the same regional shocks as the treated unit (good comovement)...
- ...but nearby units are exactly those most likely to absorb *spillovers*.

Handle it by:

- **Design**: drop plausibly-contaminated neighbours from the donor pool; or
- **Analysis**: keep them, but reason about the *direction* of the resulting bias (as in the reunification example).

Contextual requirements (5): the convex-hull condition

SCM works only if the treated unit's characteristics fall *near* the convex hull of the donors:

$(X_{11}, \dots, X_{k1})$ close to the convex hull of $\{X_2, \dots, X_{J+1}\}$.

- If the treated unit is **extreme** on some variable, no weighted average can reach it (recall West German inflation).
- Worse: if the treated unit is extreme in *pre-period outcomes*, you can't even track its outcome path.

Transformations that can help: work in *first differences* ΔY_{jt} , *growth rates*, or *deviations from pre-period means* $\tilde{Y}_{jt} = Y_{jt} - \frac{1}{T_0} \sum_h Y_{jh}$.

Caveat: differencing can inflate the noise variance (if ε_{jt} is roughly independent over time), raising overfitting risk. Reproducing the *level* is sometimes essential (e.g. growth-convergence settings).

Contextual requirements (6): time horizon

Some effects take time to emerge or to grow large enough to detect.

- Passive approach: *wait* until the effect plays out (often impractical).
- Proactive approach: use **surrogate outcomes** or **leading indicators** that move earlier.

The flip side: some effects appear strongly then *dissipate*. Enough post-periods are needed to see the full time-path, not just the first response.

Data requirements

1. Aggregate data on outcomes and predictors for the treated unit and donors.

- Usually official series (FBI crime rates, World Bank GDP, cigarette-sales statistics).
- Where no aggregate exists, aggregate up from micro-data (e.g. CPS for city-level wages, à la Mariel).

2. Sufficient pre-intervention information.

- Bias falls with T_0 — collect a long pre-period.
- Caveat: **structural breaks**. A long window may straddle a regime change (factor loadings shift). Up-weight more recent pre-periods via v_h to guard against instability.
- With few pre-periods, perfect fit can be *spurious*; adding strong predictors reduces residual variance and overfitting.

3. Sufficient post-intervention information, on outcomes relevant to the decision, to capture the effect's full time-path.

Inference: what is the right thought experiment?

Classical (sampling-based) inference imagines drawing new samples from a population. In SCM that is awkward:

- often **no well-defined sampling mechanism** — the units aren't a random draw;
- often the **sample is the population** (e.g. all Spanish regions, all OECD countries).

So SCM uses **design-based / permutation inference**: hold the data fixed and ask how *extreme* the treated unit's estimate is, relative to what we'd get if the *treatment label* were on some other unit.

When treatment is genuinely randomized this reduces to Fisher's (1935) randomization inference — but random assignment is rare for aggregate interventions.

Placebo (permutation) tests

Procedure:

1. For each donor j , pretend it was treated; build its synthetic control from the remaining units; estimate a “placebo gap” $Y_{jt} - \hat{Y}_{jt}^N$.
2. Pool the real treated gap with all placebo gaps.
3. The effect is significant if the real gap is **extreme** relative to the placebo distribution.

This is feasible *only because* SCM gives a well-defined, reapplicable rule for choosing a comparison — you can run the exact same procedure on every donor.

Fixing the “bad pre-fit” problem: the RMSPE ratio

A placebo donor that SCM fits *badly* before treatment will show a big post-gap for no good reason. So normalize by pre-fit.

Root mean squared prediction error over periods $t_1, \dots, t_2$:

$$R_j(t_1, t_2) = \left(\frac{1}{t_2 - t_1 + 1} \sum_{t=t_1}^{t_2} (Y_{jt} - \hat{Y}_{jt}^N)^2 \right)^{1/2}.$$

Test statistic = **post/pre ratio**:

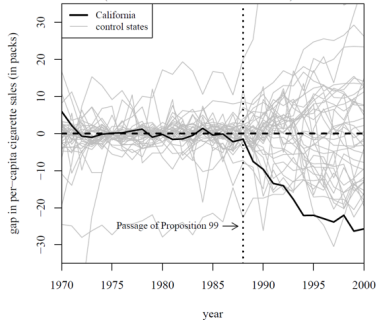
$$r_j = \frac{R_j(T_0 + 1, T)}{R_j(1, T_0)}.$$

Large r_j = big post-period divergence *relative to* how well the unit was fit beforehand. This is the quantity we permute.

Placebo gaps for every state

Application: California tobacco control program

(ALL STATES IN DONOR POOL)

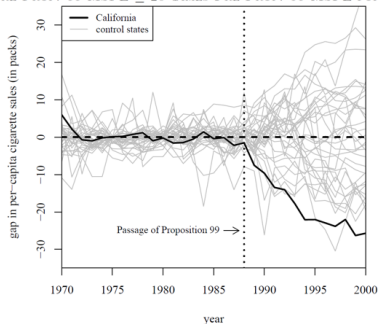


California's post-period gap is large — but so are a few placebos that were *poorly fit* before 1988. We need to filter those out.

Filtering by pre-period fit

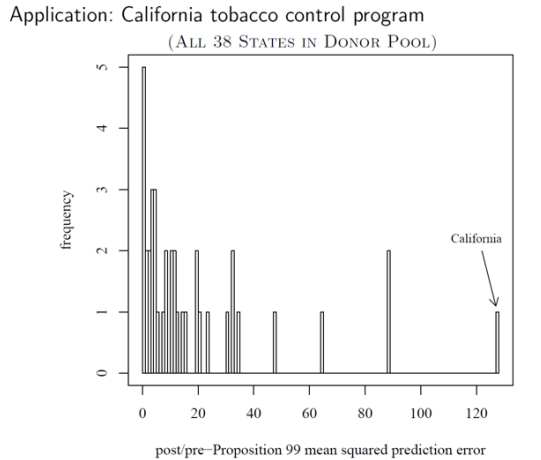
Application: California tobacco control program

(PRE-PROP. 99 MSPE $\leq$ 20 TIMES PRE-PROP. 99 MSPE FOR CA)



Restricting to placebos with pre-period fit comparable to California's (here, $\text{MSPE} \leq 2\times$): California's gap is now clearly the most extreme negative path.

The ratio distribution



California's r_j sits in the extreme right tail. With 38 states, being the most extreme gives a permutation p -value $\approx 1/39 \approx 0.026$.

p -values, one-sided tests, and what to report

Permutation p -value (rank of the treated statistic):

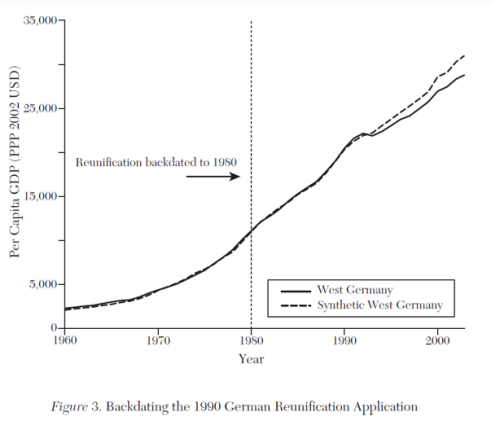
$$p = \frac{1}{J+1} \sum_{j=1}^{J+1} \mathbf{1}\{r_j \geq r_1\}.$$

- **One-sided** versions (using positive or negative parts of the gap) can gain substantial power — valuable given tiny samples.
- **Confidence sets** by test inversion (Firpo & Possebom 2018).
- Report the **whole permutation distribution** (gap plots, ratio histogram), not just a p -value — it shows the *magnitude* of the treated effect against placebos.

The benchmark is usually a *uniform* assignment distribution; sensitivity analyses can relax this.

Robustness 1 — Backdating / in-time placebo

Re-estimate pretending the intervention happened **earlier** than it did.



Good signs: (i) no gap appears in the fake post-period before the *real* event (the “pre-program test”); (ii) the gap still appears around the true date. Credibility for the design.

Robustness 2 — Leave-one-out

Drop each contributing donor in turn and re-estimate.

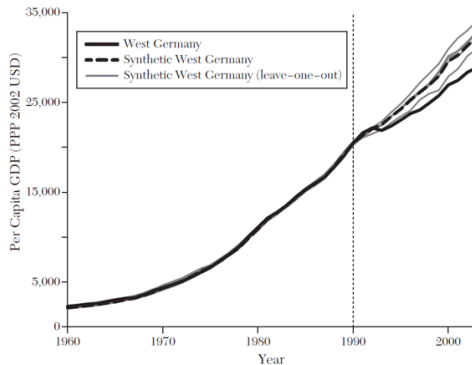


Figure 4. Leave-one-out Estimates of the Effect of the 1990 German Reunification

If the conclusion survives removing any single donor (as here — all paths negative, centred on the main estimate), no one unit is driving the result.

Robustness 3 — Sensitivity to design choices

Two design levers shape every SCM result:

- the **donor pool** (columns of X_0), and
- the **predictors** (rows of $[X_1 : X_0]$).

Good practice: show the main conclusion is stable across reasonable alternatives.

Diagnostic flag: if removing a donor changes the estimate a lot *without* hurting pre-period fit, investigate — the excluded unit may have had its own intervention or a large idiosyncratic shock.

The overarching rule

SCM is credible only when the synthetic control reproduces the treated unit's pre-period *outcome path closely*. No close fit $\Rightarrow$ no SCM.

Extensions

Map of the extensions

The canonical estimator has been extended along several lines. We cover the main ideas (inference kept light):

- **Penalized SC** — uniqueness, less interpolation bias.
- **Augmented / bias-corrected SC** — fix imperfect pre-fit with a regression adjustment.
- **Regression-relaxed weights** — elastic net, level shifts (Doudchenko & Imbens).
- **Synthetic difference-in-differences** — weight units *and* time periods.
- **Multiple & staggered treated units.**
- **Matrix completion** — impute the missing counterfactual.
- **SCM for experimental design** — the A/B-testing link.

Penalized synthetic control (Abadie & L'Hour)

Add a penalty on *pairwise* distances between the treated unit and each donor it uses:

$$\min_w \left\| X_1 - \sum_j w_j X_j \right\|^2 + \lambda \sum_j w_j \|X_1 - X_j\|^2 \quad \text{s.t. } w_j \geq 0, \sum_j w_j = 1.$$

λ trades off two goals:

- $\lambda \rightarrow 0$: the *pure* synthetic control (best aggregate fit; among ties, least pairwise discrepancy).
- $\lambda \rightarrow \infty$: *nearest-neighbour matching* (one closest donor).

“Aggregate fit” can be achieved by averaging a far-left and a far-right donor — a poor match to each. The penalty discourages using donors that are individually far from the treated unit.

Penalized SC — why it helps, and a 1-D example

Advantages: for any $\lambda > 0$ the solution is **unique and sparse** (under mild conditions); it **reduces interpolation bias**; same computational cost as the unpenalized estimator.

$X_1 = 2$, donors at 1, 4, 5.



- Unpenalized: *two* sparse perfect fits, e.g. $(\frac{2}{3}, \frac{1}{3}, 0)$ using $\{1, 4\}$ and $(\frac{3}{4}, 0, \frac{1}{4})$ using $\{1, 5\}$ — plus infinitely many mixtures.
- Penalized ($\lambda > 0$): *unique*, and never uses the far donor $X_4 = 5$.
- As $\lambda \rightarrow 0$ it selects the better-matched $\{1, 4\}$ solution.

Augmented / bias-corrected SC

When the synthetic control *cannot* match X_1 exactly, the residual imbalance biases $\hat{\tau}$. Correct it with a regression, à la bias-corrected matching:

$$\hat{\tau}_{1t} = \underbrace{\left(Y_{1t} - \sum_j w_j^* Y_{jt} \right)}_{\text{SCM estimate}} - \underbrace{\sum_j w_j^* (\hat{\mu}_{0t}(X_1) - \hat{\mu}_{0t}(X_j))}_{\text{adjust for the imbalance } X_1 - X_0 W^*}.$$

where $\hat{\mu}_{0t}(\cdot)$ regresses donor outcomes on their predictors. Equivalently: **apply SCM to regression residuals**.

- Ben-Michael, Feller & Rothstein (2021): the **Augmented SCM (ASCM)** — ridge-augmented weights that allow a small, controlled amount of extrapolation to improve pre-fit, with a penalty keeping it close to honest SCM.

Most useful when good fit is otherwise unattainable (e.g. many treated units, or a treated unit on the hull boundary).

Regression-relaxed weights (Doudchenko & Imbens)

Relax SCM's constraints and regularize instead:

$$\hat{Y}_{1t}^N = \hat{\alpha} + \sum_j \hat{w}_j Y_{jt},$$

with (α, w) minimizing pre-period fit plus an **elastic-net** penalty (mix of lasso L_1 and ridge L_2):

- allows an **intercept shift** α (levels can differ), like DiD;
- allows **negative weights** / sums $\neq 1$ (extrapolation), but the penalty controls them;
- nests several methods: pure SC, constrained regression (Hsiao et al.), lasso-SC.

Trade-off: more flexibility and often better fit, at the cost of SCM's clean “convex average of named units” interpretation.

Synthetic difference-in-differences (Arkhangelsky et al.)

SDID blends DiD and SCM by weighting in *two* dimensions:

- **Unit weights** $\hat{\omega}_j$ — like SCM, make donors track the treated unit's pre-period *level/trend*.
- **Time weights** $\hat{\lambda}_t$ — up-weight pre-periods that resemble the post-period.

Then run a *weighted* two-way fixed-effects DiD.

- DiD: all units, all periods equal $\Rightarrow$ relies on parallel trends.
- SCM: unit weights only, fit levels.
- SDID: unit *and* time weights $\Rightarrow$ robust to violations of strict parallel trends; allows a level offset (doesn't force exact pre-period fit).

Often more stable than SCM when pre-fit is imperfect, with a well-developed inference theory.

Multiple & staggered treated units

Multiple treated units, same timing:

- Conceptually easy: fit a *separate* synthetic control for each treated unit, then average effects (possibly population- or GDP-weighted).
- Practical snag: with many treated units, some X_i fall *inside* the donor hull $\Rightarrow$ non-unique/non-sparse weights, and large interpolation bias hidden under “perfect fit.” The penalized estimator restores uniqueness.

Staggered adoption (units treated at *different* times):

- For each treated unit, use as donors only units *not-yet-treated* (or never-treated) at the relevant time; build its synthetic control on its own pre-period; align by *event time*.
- Aggregate the unit-by-unit, time-since-treatment effects — closely related to modern staggered-DiD estimators and to **SDID for staggered designs**.

This is the active frontier where SCM meets the staggered-DiD literature.

Matrix completion

View the panel of no-treatment outcomes $\{Y_{jt}^N\}$ as a **matrix with missing entries** — the treated unit's post-period cells are exactly the unknown counterfactuals.

- Assume the underlying matrix is **low-rank** (a few latent factors generate it).
- Use matrix-completion / singular-value-thresholding to impute the missing cells.
- The imputed treated post-period values are the counterfactual $\hat{Y}_{1t}^N$.

Generalizes SCM: uses information from *all* cells (not just a convex average of donors), de-noises the data, and naturally handles missing values and multiple treated units. Extensions add covariates and fixed effects.

SCM for experimental design — the A/B-testing link

Turn SCM around: use it to *design* an experiment when you can treat only one/few units (markets, cities).

Problem (e.g. a ridesharing firm piloting a new driver-incentive in a few cities): randomization can produce a *defective* design — the treated cities may be unrepresentative or very different from the controls.

Abadie & Zhao (2021): use *two* synthetic units:

- a synthetic **treated** unit chosen to represent the whole population of markets, and
- a synthetic **control** for that synthetic treated unit.

This is why business analytics teams (Uber, Microsoft/Xandr) use SCM as an *experimental* tool, not just an observational one — it sidesteps costly, sometimes infeasible, full A/B tests.

Choosing among the variants — practical guidance

- **One treated unit, good pre-fit, smooth outcome** $\Rightarrow$ canonical SCM. Start here.
- **Imperfect pre-fit** $\Rightarrow$ augmented SCM or synthetic DiD (allow a controlled offset).
- **Many / micro-level treated units, non-unique weights** $\Rightarrow$ penalized SCM.
- **Want DiD-style robustness with data-driven weights** $\Rightarrow$ synthetic DiD.
- **Staggered timing** $\Rightarrow$ event-time SCM / staggered SDID.
- **Designing a pilot you control** $\Rightarrow$ SCM for experimental design.

Whatever the variant: the credibility argument is the same — close, honest pre-period fit, a clean donor pool, and transparent reporting.

Wrap-up

Key takeaways

1. SCM estimates the effect of an intervention on *one* aggregate unit by building a **convex-weighted synthetic control** from a donor pool.
2. The effect is the **post-period gap** between the treated unit and its synthetic control.
3. Its edge over regression: **no extrapolation, transparent fit, no specification search, interpretable + sparse weights.**
4. Validity rests on **close pre-period fit**, a **clean donor pool**, no anticipation, no interference, and the convex-hull condition.
5. Inference is **design-based** (placebo/permutation), since there is no sampling mechanism.
6. Always run **diagnostics**: backdating, leave-one-out, sensitivity to donors/predictors.

When is SCM the right tool? A checklist

Use SCM when:

- there is **one** (or a few) treated aggregate unit(s);
- a **sharp, datable** intervention;
- several plausible **donor** units, none contaminated by the same treatment;
- a **long pre-period** and a reasonably **smooth** outcome;
- the treated unit is **not extreme** (sits near the donors' convex hull);
- the expected effect is **large relative to noise**.

If close pre-period fit is unattainable, **do not force SCM** — transform the outcome, switch to SDID/augmented SCM, or reconsider the design.

Open research: sampling-based inference; external validity; sensitivity to model restrictions; multiple/staggered interventions; mechanisms/mediation; data-driven choice of predictor weights v_h ; fast, robust computation.

Software:

- `Synth` (R, Stata, MATLAB) — the canonical estimator.
- `pensynth` (R) — penalized SC.
- `augsynth` (R) — augmented SCM & staggered adoption.
- `synthdid` (R) — synthetic difference-in-differences.
- `tidysynth` (R), `scpi` (R/Stata/Python) — modern SCM with prediction intervals.
- Python: `SparseSC`, `causalimpact`-style tools.

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Questions?